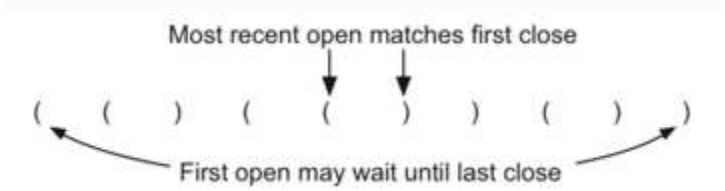
Balanced Parentheses

(5+6)∗(7+8)/(4+3) balanced
2 * ((4/2) + 5 not balanced

```
(((()[]()()))

{((()))}

(()(((())()))
```



Algorithm:

- Declare a character stack S.
- Now traverse the expression string exp.
  1. If the current character is a starting bracket (**'(' or '{' or '['**) then push it to stack.
  2. If the current character is a closing bracket (**')' or '}' or ']'**) then pop from stack and if the popped character is the matching starting bracket then fine else brackets are not balanced.
- After complete traversal, if there is some starting bracket left in stack then "not balanced"